Theodolite

Theodolite is an instrument which is used in measuring the angle between an object and the eye of the observer. A theodolite consists of two graduated wheels placed at right angles to each other and a telescope. The wheels are used for the measurement of horizontal and vertical angles. The angle to the desired point is measured by positioning the telescope towards that point. The angle can be read on the telescope scale.



Fig. 6.6

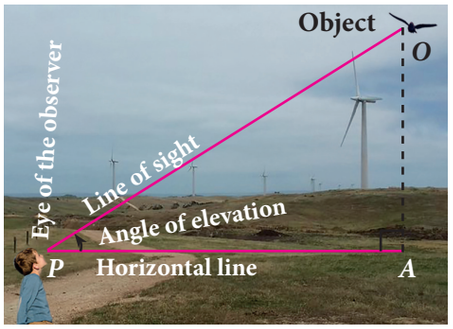


Fig. 6.7

Angle of Elevation

The angle of elevation is an angle formed by the line of sight with the horizontal when the point being viewed is above the horizontal level. That is, the case when we raise our head to look at the object. (see Fig. 6.7)

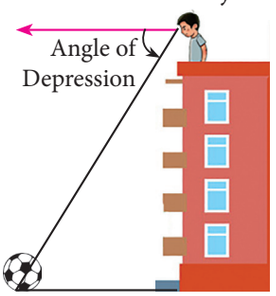


Fig. 6.8

Angle of Depression

The angle of depression is an angle formed by the line of sight with the horizontal when the point is below the horizontal level. That is, the case when we lower our head to look at the point being viewed. (see Fig. 6.8)

Clinometer

The angle of elevation and depression are usually measured by a device called clinometer.

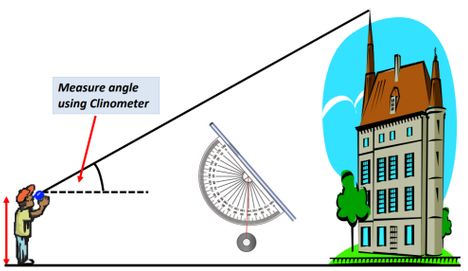


Fig. 6.9

Note

- From a given point, when height of an object increases the angle of elevation increases.

  If $h_1 > h_2$ then $\alpha > \beta$

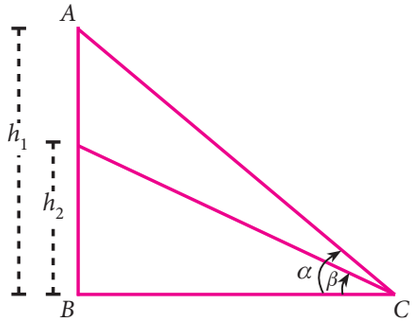


Fig. 6.10(a)

- The angle of elevation increases as we move towards the foot of the vertical object like tower or building.

  If $d_2 < d_1$ then $\beta > \alpha$

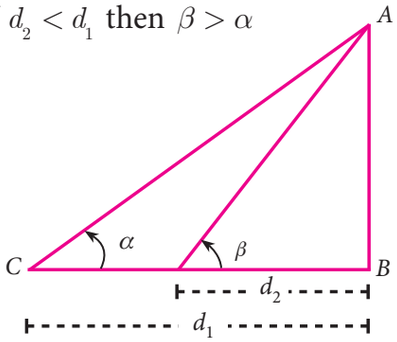


Fig. 6.10(b)

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